Fast Fourier Transform Algorithm

Let $Y_k = \sum_n X_n \cdot e^{-\frac{2\pi i}{N}nk}$ where $X_n, Y_k \in \mathbb{C}$ for $n, k \in \{\,0,\ ...,\ N-1\,\}$ with time complexity $O(N^2)$

Hence, $Y_k = \sum_m X_{2m} \cdot e^{-\frac{2\pi i}{N}(2m)k} + \sum_m X_{2m+1} \cdot e^{-\frac{2\pi i}{N}(2m+1)k}$ for $m \in \{\,0,\ ...,\ \frac{N}{2}-1\,\}$ and $k \in \{\,0\,,\ ...,\ N-1\,\}$

Hence, $Y_k = \sum_m X_{2m} \cdot e^{-\frac{2\pi i}{N/2}mk} + e^{-\frac{2\pi i}{N}k} \cdot \sum_m X_{2m+1} \cdot e^{-\frac{2\pi i}{N/2}mk}$ for $m \in \{\,0,\ ...,\ \frac{N}{2}-1\,\}$ and $k \in \{\,0\,,\ ...,\ \frac{N}{2}-1\,\}$

Hence, $Y_k = \sum_m X_{2m} \cdot e^{-\frac{2\pi i}{N/2}mk} - e^{-\frac{2\pi i}{N}k} \cdot \sum_m X_{2m+1} \cdot e^{-\frac{2\pi i}{N/2}mk}$ for $m \in \{\,0,\ ...,\ \frac{N}{2}-1\,\}$ and $k \in \{\,\frac{N}{2},\ ...,\ N-1\,\}$

Let $E_k = \sum_m X_{2m} \cdot e^{-\frac{2\pi i}{N/2}mk}$, $O_k = \sum_m X_{2m+1} \cdot e^{-\frac{2\pi i}{N/2}mk}$, $\omega = e^{-\frac{2\pi i}{N}}$

Hence, $Y_k = E_k + \omega^k \cdot O_k$ for $k \in \{\,0\,,\ ...,\ \frac{N}{2}-1\,\}$

Hence, $Y_k = E_k - \omega^k \cdot O_k$ for $k \in \{\,\frac{N}{2},\ ...,\ N-1\,\}$ with time complexity $O(N \log N)$